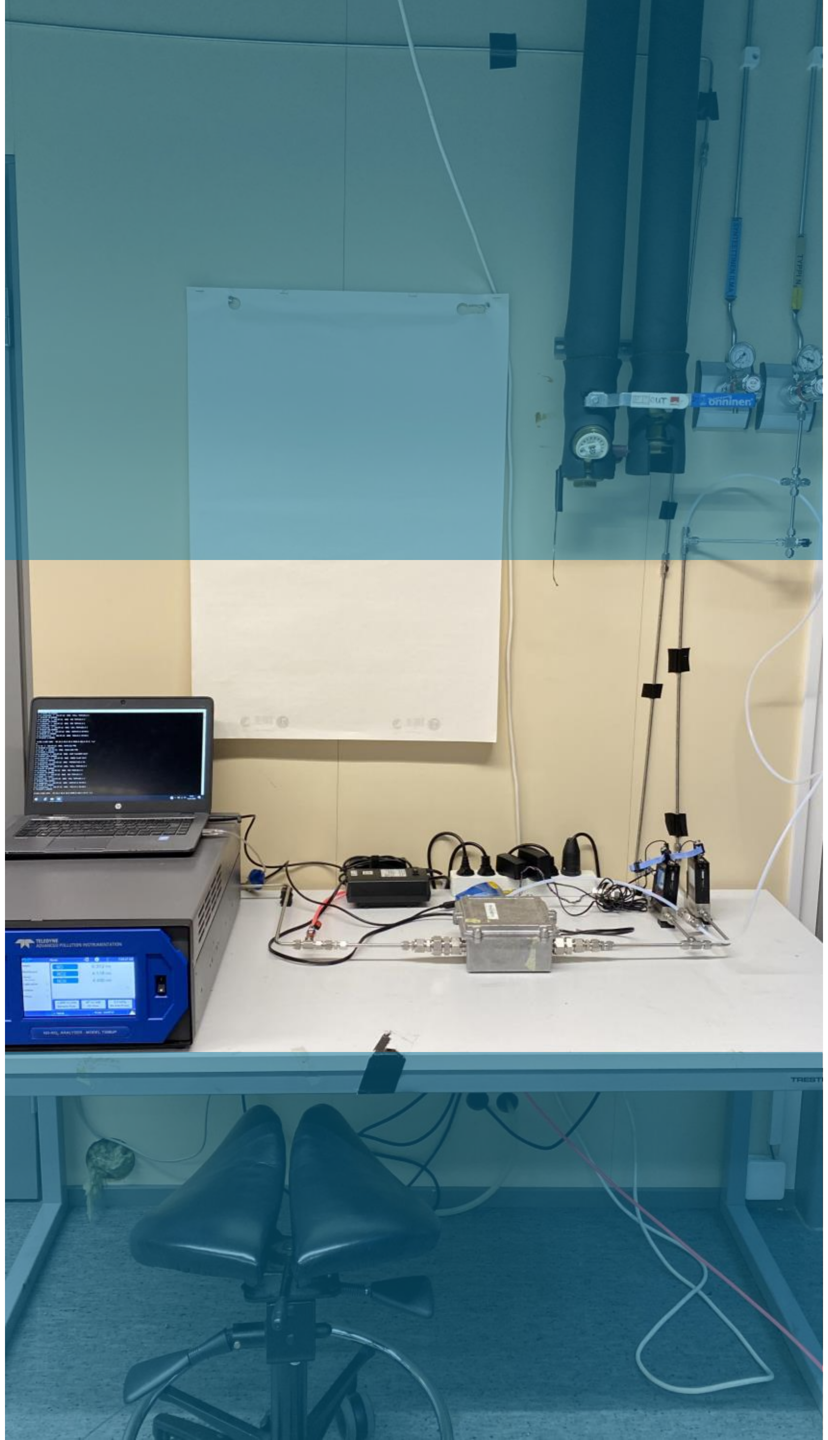
N_2O Actinometry Experiment

Operational instructions for the ACTRIS CiGas – UHEL setup



Cecilia Righi, Nina Sarnela

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Outline

Theoretical background

Calibration experiment

- Experimental setup
- Before the experiment
- Performing the experiment
- H2SO4-calibration-Alicat

Calculate It-product

- Option A
- Option B
- Notes

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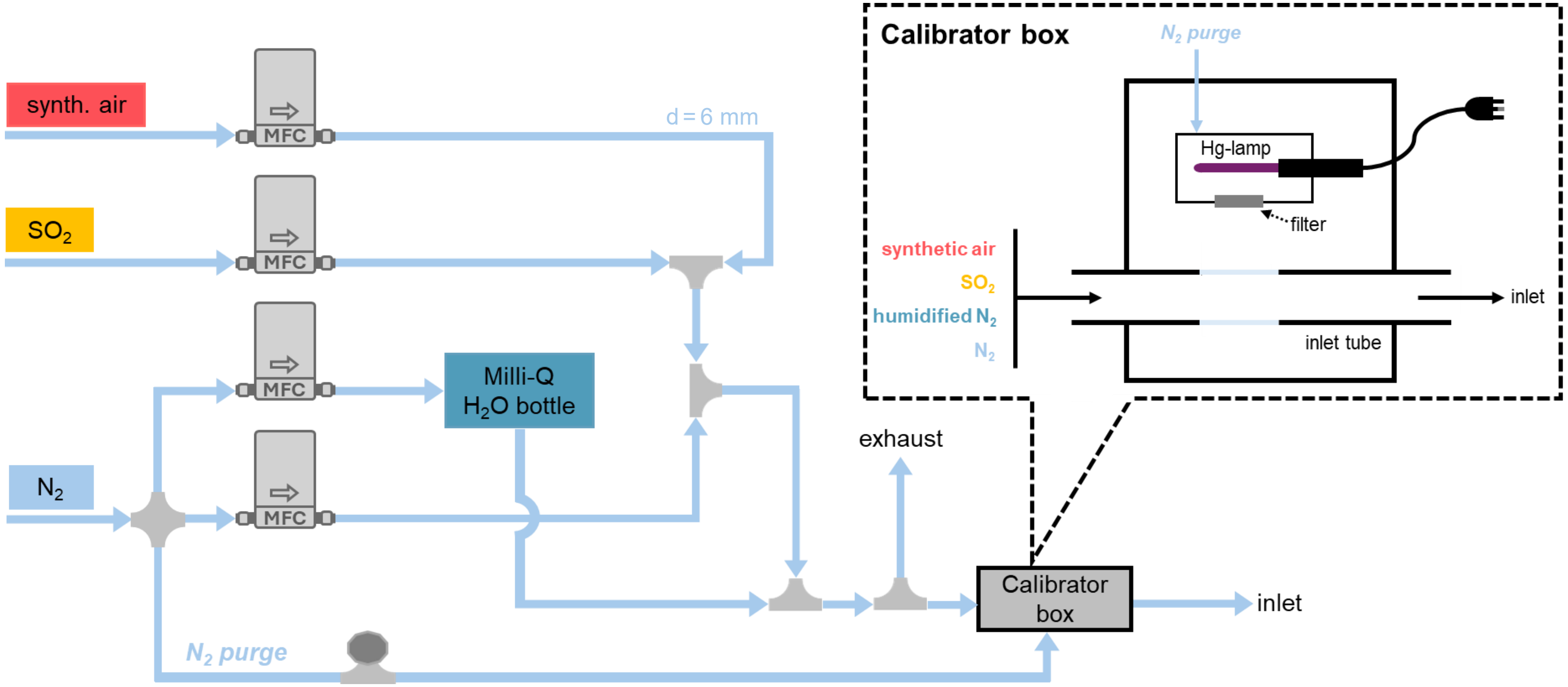
Theoretical background

❖ The sulfuric acid calibration method for chemical ionization mass spectrometers described by Kürten et al. (2012) includes a step in which the **theoretical sulfuric acid concentration ([H_2SO_4])** introduced into the instrument at each stage of the calibration experiment must be modeled. Within this model, [H_2SO_4] is estimated based on the **theoretical hydroxyl radical concentration ([OH])** generated in the calibrator box, where a Pen-Ray® Hg lamp (Analytik Jena) is installed



$$H_2O + h\nu_{(184.9\,nm)} \rightarrow \mathbf{OH} + H$$

$$SO_2 + \mathbf{OH} \rightarrow HSO_3$$

$$HSO_3 + O_2 \rightarrow SO_3 + HO_2$$

$$SO_3 + 2H_2O \rightarrow H_2SO_4 + H_2O$$

Theoretical background

❖ The Hg lamp, emitting UV light at 184.9 nm, is used to initiate the **photolysis of water vapor** (**H_2O**), thereby generating OH radicals. The general equation used to calculate the OH concentration produced by this photolysis reaction is given by:

$$[OH] = I \cdot t \cdot \sigma_{H_2O} \cdot \Phi_{H_2O} \cdot [H_2O] \quad \text{(eq. 1)}$$

where I = photon intensity

t = illumination time

σ_{H_2O} = absorption cross section of H_2O vapor at 184.9 nm

Φ_{H_2O} = quantum yield of H_2O photodissociation

$[H_2O]$ = concentration of water vapor

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Theoretical background

- σ_{H_2O} and Φ_{H_2O} have tabulated values of 7.22 × 10^{-20} cm^2 and 1.0, respectively. [H_2O], instead, is calculated by the model according to Equation 2:

$$[H_2O] = \frac{Q_{H_2O}}{Q_{N_2}+Q_{H_2O}+Q_{SO_2}+Q_{air}} \cdot \frac{p_{sat}(T)\cdot N_A}{R\cdot T} \qquad \text{(eq. 2)}$$

where T = temperature

p_{sat}= saturation vapor pressure of water

Q_x = flow rates of the gas x introduced into the calibration system

- Hence, $\boldsymbol{\sigma_{H_2O}}$, $\boldsymbol{\Phi_{H_2O}}$, **and** $[\boldsymbol{H_2O}]$ **for each step in Equation 1 are known, whereas** $\boldsymbol{I}$ **and** $\boldsymbol{t}$ **are determined through chemical actinometry**. Specifically, they are determined as their combined product, hereafter referred to as the ***It-product***.

Theoretical background

- In the present case, this involves using a separate chemical reaction with known reaction rates and accurately measurable products in an experiment conducted under the same conditions as the sulfuric acid calibration experiment.

- The reaction selected for the actinometry experiment is the **conversion of nitrous oxide (N_2O) into nitrogen oxides (NO_x)**, as described by Edwards et al. (2003):

$$\mathrm{N_2O + h\nu\ (\lambda = 184.9\ nm) \rightarrow N_2 + O(^1D)} \quad \text{(R1)}$$

$$\mathrm{O(^1D) + N_2 \rightarrow O(^3P) + N_2} \quad \text{(R2)}$$

$$\mathrm{O(^1D) + N_2O \rightarrow 2NO} \quad \text{(R3)}$$

$$\mathrm{O(^1D) + N_2O \rightarrow N_2 + O_2} \quad \text{(R4)}$$

$$\mathrm{O(^1D) + N_2O + M \rightarrow O(^3P) + N_2O} \quad \text{(R5)}$$

$$\mathrm{O(^3P) + O_2 + M \rightarrow O_3 + M} \quad \text{(R6)}$$

Theoretical background

- The NO produced by reaction R3 is typically 5 ppbv or less and is measured using a commercial **NO_x detector**. Ozone formation via reaction R6 may pose a potential issue, as O_3 can remove NO from the system by oxidizing it to NO_2. However, according to Edwards et al. (2003), NO accounts for more than 80% of the total [NO_x].

- The relationship between the *It-product* and the concentrations of NO_x, N_2O, and N_2 is given by Equation 3:

$$It_{N_2O} = \frac{k_2 \cdot [N_2] + (k_3 + k_4 + k_5) \cdot [N_2O]}{2k_3 \cdot \sigma_{N_2O} \cdot \varphi_{N_2O} \cdot [N_2O]^2} \cdot [NO_x] \qquad \text{(eq. 3)}$$

Table 1 *Absorption cross section, photochemical yield, and reaction rate values used to calculate the It-product for the conversion of N_2O to NO_x.*

reaction and parameter	value or expression
R1, σ_{N2O} (cm^2)	1.43×10^{-19}
R1, Φ_{N2O} (cm^2)	1
R2, k_2 ($cm^3\ mol^{-1}\ s^{-1}$)	2.0×10^{-11} exp(130/T(K))
R3, k_3 ($cm^3\ mol^{-1}\ s^{-1}$)	7.6×10^{-11}
R4, k_4 ($cm^3\ mol^{-1}\ s^{-1}$)	4.3×10^{-11}
R5, k_5 ($cm^3\ mol^{-1}\ s^{-1}$)	6.0×10^{-12}

Theoretical background

- As detailed in Appendix A of Kürten et al. (2012), attenuation of UV light along its path through the quartz tube is non-negligible, even at the low N_2O/N_2 mixing ratios used in the actinometry experiment. Therefore, a **geometry-dependent correction factor** must be calculated. According to the authors, the geometry factor $\boldsymbol{K}$, which depends on both the N_2O concentration and the tube dimensions, is computed as

$$K = \frac{2\int_{r=0}^{R}\int_{\varphi=0}^{\pi} \exp\left(-\sigma_{N_2O}\cdot[N_2O]\cdot\left(\sqrt{R^2-(r\cdot\sin\varphi)^2}+r\cdot\cos\varphi\right)\right)\cdot r\cdot d\varphi\cdot dr}{\pi\cdot R^2} \qquad \text{(eq. 4)}$$

where $R\ (cm)$ = radius of the quartz tube

$(\sqrt{R^2-(r\cdot\sin\varphi)^2}+r\cdot\cos\varphi)$ = absorption path length

- The geometry factor K must be included when calculating It_{N_2O}:

$$It_{N_2O\_corrected} = \frac{k_2\cdot[N_2]+(k_3+k_4+k_5)\cdot[N_2O]}{2k_3\cdot\sigma_{N_2O}\cdot\varphi_{N_2O}\cdot[N_2O]^2}\cdot\frac{[NO_x]}{K} \qquad \text{(eq. 5)}$$

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Theoretical background

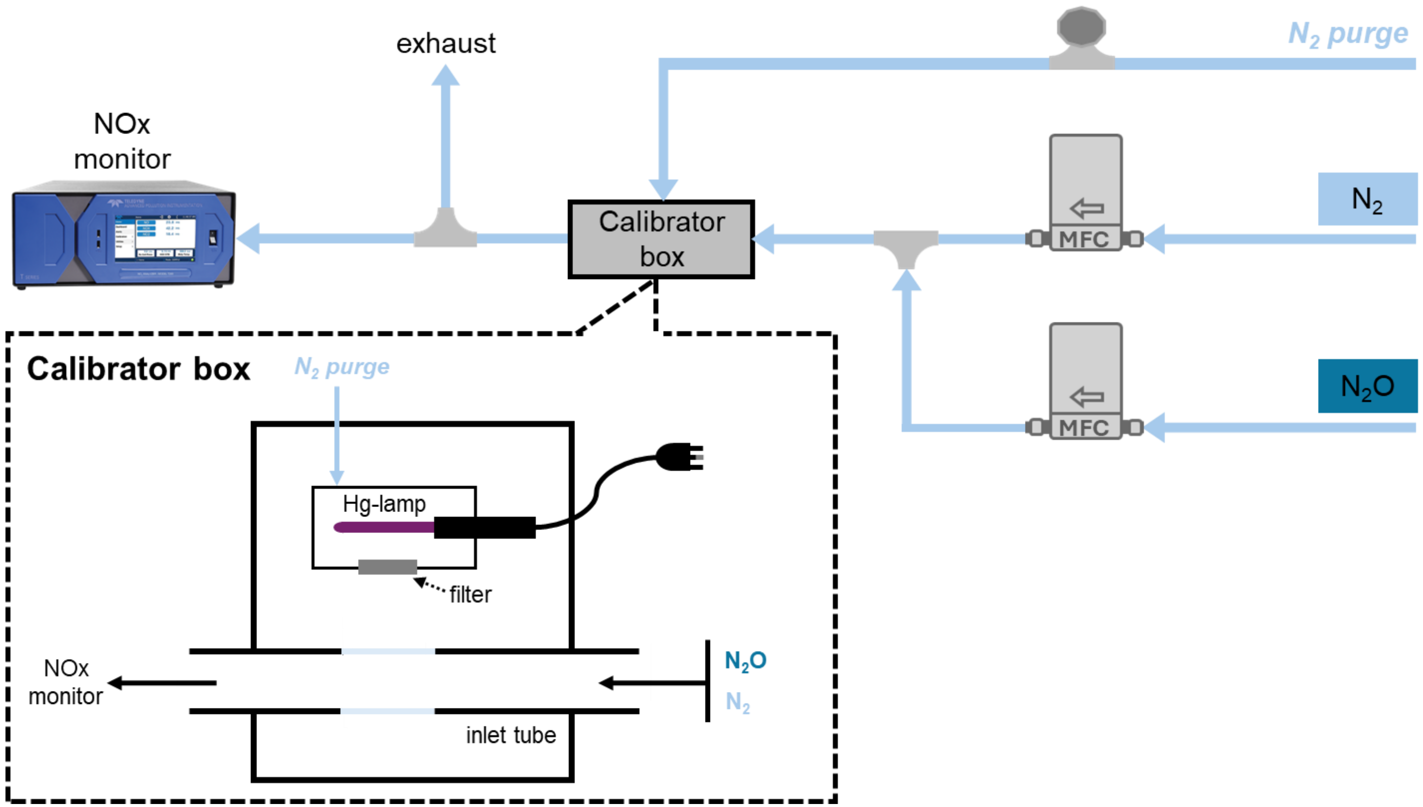
❖ Finally, Kürten et al. (2012) demonstrated that different flow rates can be used during the actinometry and sulfuric acid calibration experiments. The derived ***It-product*** can then be scaled using the flow-rate ratio according to Equation 6:

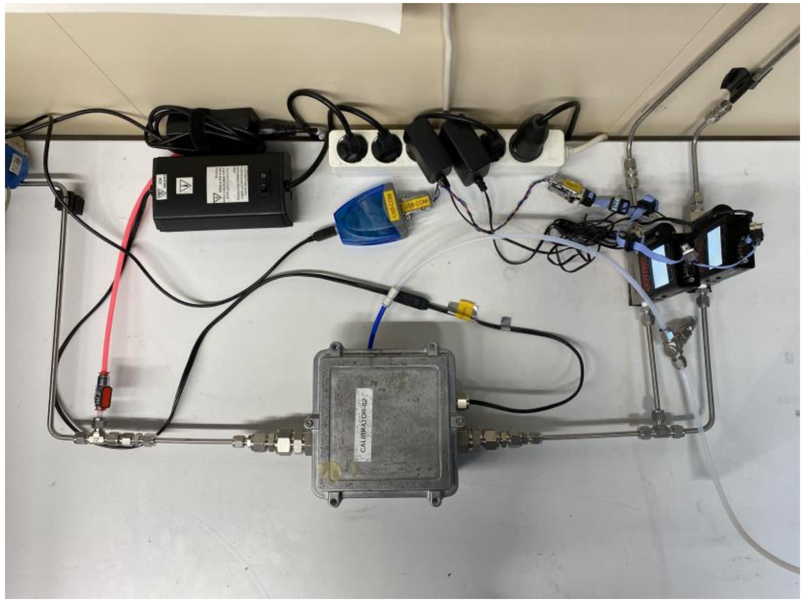
$$It = \frac{Q_{total,\,N_2O}}{Q_{total,H_2SO_4}} \cdot It_{N_2O\_corrected} \cdot \qquad \text{(eq. 6)}$$

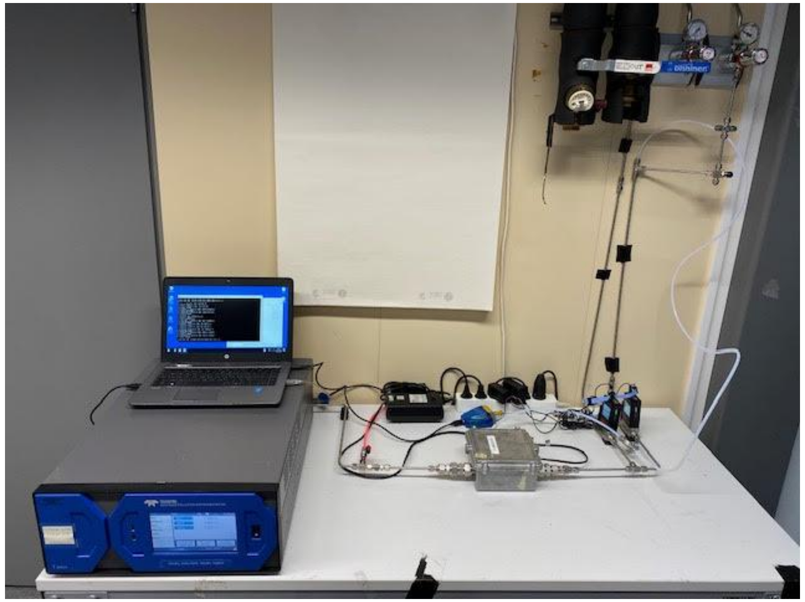
where $Q_{total,\,N_2O}$ = total flow used for the N_2O actinometry experiment
Q_{total,H_2SO_4} = total flow used for the H_2SO_4 calibration experiment

Calibration experiment

Experimental setup







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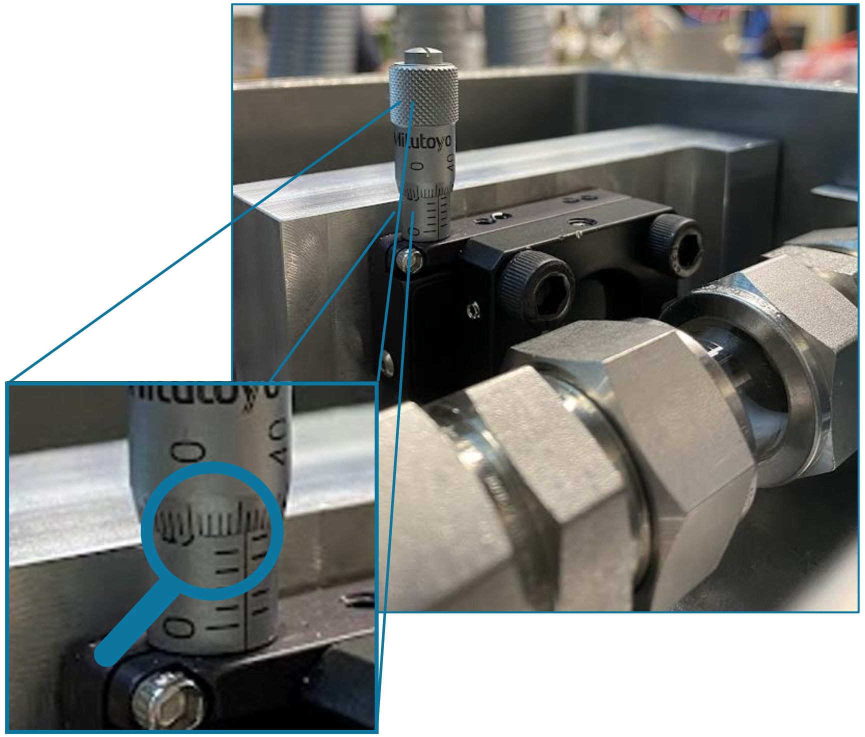
Calibration experiment

Before the experiment

❖ The NO_x monitor used for the experiment must be powered on at least two days in advance to ensure stable operation before the actinometry experiment begins. During this period, allow it to sample room air (make sure it is not sampling from the exhaust line).

❖ Check that there are no cracks in the quartz tube in the calibrator.

❖ Ensure that the filter in the calibrator is set to the same position used during the sulfuric acid calibration experiment (5 mm for this setup).



Calibration experiment

Performing the experiment

- ❖ Place the calibrator in position and connect the setup to the NO_x monitor.
- ❖ Set the N_2 flush to the same flow rate used during the calibration experiment (1.5 lpm).
- ❖ Open *rslog.exe* (cd *C:\Users\mittaaja\Desktop\gaslog*) to start recording NO and NO_2 measurements.
- ❖ Start the background experiment by setting the mass flow controllers to the same N_2O and N_2 flow rates that will be used during the actinometry experiment; a minimum of 10 steps, each lasting 10 minutes, is recommended. The flow rates can be set either manually or by running the dedicated Python script (see Slide 13).
- ❖ Turn on the Hg lamp and allow it to warm up for the same duration as during the sulfuric acid calibration experiment (~1.5 hours for calibrators 1, 3, and 4; ~2 hours for calibrator 2).
- ❖ Once the Hg lamp is fully warmed up, start the actinometry experiment by setting the mass flow controllers to the desired N_2O and N_2 flow rates. Make sure to also measure the background signal with no N_2O introduced.

Calibration experiment

H2SO4-calibration-Alicat

1. Adjust the flow settings as needed in *config.ini*

- .vscode
- __pycache__
- data
- debug
- alicat_mfc.py
- automatic_measurement.py
- config.ini
- config.py
- directory_list.txt
- main.py
- Time

```
# General notes on editing this file:
# - Comments must be on their own line. E.g. 'device_id = Dev1 # Qwerty' doesn't work (Maybe there is a way to get it working)
# - Section names probably allows whitespace but I wanted to avoid any potential problems by the naming convention currently used


# MFC's serial port settings
[MFC:Serial_port]
port = COM10
baudrate = 19200
bytesize = 8
parity = N
stopbits = 1
timeout = 1
# Software flow control
xonxoff = 0
# Hardware flow control
rtscts = 0
# individual addresses. NOTE: each MFC has individual address! (A = N2, B = N2O)
address = A,B

# Scaling settings for the MFCs
[MFC:Scaling]
# Flow
f_span = 1,1
f_offset = 0,0
# Temperature
t_span = 1,1
t_offset = 0,0
# Pressure
p_span = 68.9476,68.9476
p_offset = 0,0

# Automatic measurement loop settings
[Automatic_measurement]
# Measurement frequency [Hz]
freq = 1

# Set time intervals (in seconds) to change the flow rates. NOTE: these are indeed time intervals Delta_t, not time from starting the script.
# time_intervals = 600,600,600,600,600,600,600,600,600,600,60
# Setting flow rate [sLPM or sccm, depends on the MFC] for each MFC. Symbol "--" separates different time intervals.
# set_flow_init = 6.696,0.804--6.466,1.034--6.250,1.250--6.048,1.452--5.859,1.641--5.682,1.818--5.515,1.985--5.357,2.143--5.208,2.292--5.068,2.432--0,0
```

COM port corresponding to the connection to the MFCs (check in *Device Manager*)

Verify and, if needed, correct the individual MFC addresses shown on the MFC displays

Set each step duration and corresponding flows following the instructions

2. Run *main.py* from the terminal

Calculate the It-product

Option A - Using example_sheet.xlsx and It_factor_correction.ipynb

- Duplicate the worksheet example in *example_sheet.xlsx* and modify the sections highlighted in green (see Slide 15-16).

- Calculate the correction factor K for each step using the Python script It_factor_correction.ipynb. This calculation must be repeated whenever the N_2O/N_2 mixing ratios in the *Actinometry experiment table* are modified (the corresponding values in the *UV light attenuation correction table* are updated automatically).

- Enter the calculated K values into the designated section of the spreadsheet (*UV light attenuation correction table*).

- The worksheet outputs the mean and median values of both the uncorrected and geometry-corrected *It-product* for the total flow rate used during the actinometry experiment (7.5 lpm), as well as for a total flow rate of 10 lpm.

example_sheet.xlsx

It_factor_correction.ipynb

Calculate the It-product

Option A - Using example_sheet.xlsx and It_factor_correction.ipynb

Info:

Date	dd/mm/yyyy
Calibrator	n
Operator	Name Surname
Email address	name.surname@helsinki.fi

It-product calculation:

parameter	value or expression
R1, σ_{N2O} (cm^2)	1.43E-19
R1, Φ_{N2O} (cm^2)	1.00E+00
R2, k_2 (cm^3 mol^{-1} s^{-1})	3.09E-11
R3, k_3 (cm^3 mol^{-1} s^{-1})	7.60E-11
R4, k_4 (cm^3 mol^{-1} s^{-1})	4.30E-11
R5, k_5 (cm^3 mol^{-1} s^{-1})	6.00E-12

T (K) 298.15

Actinometry experiment:

step	time (UTC+2)	volume mixing ratio N_2O/N_2	standard N_2O flow (sccm)	standard N_2 flow (sccm)	$[N_2O]$ (ppbv)	$[N_2]$ (ppbv)	$[NO_x]$ (ppbv)	$[NO_x]$ corrected (ppbv)	It-product (photons cm^{-2})	corrected It-product (photons cm^{-2})
1	11:51-12:00	0.12	804	6696	1.07E+08	8.93E+08	1.028	0.536	8.81E+10	1.10E+11
2	12:00-12:10	0.16	1034	6466	1.38E+08	8.62E+08	1.355	0.845	8.97E+10	1.20E+11
3	12:10-12:20	0.20	1250	6250	1.67E+08	8.33E+08	1.722	1.093	8.44E+10	1.19E+11
4	12:21-12:30	0.24	1452	6048	1.94E+08	8.06E+08	2.036	1.408	8.50E+10	1.27E+11
5	12:30-12:40	0.28	1641	5859	2.19E+08	7.81E+08	2.463	1.825	9.04E+10	1.41E+11
6	12:40-12:50	0.32	1818	5682	2.42E+08	7.58E+08	2.640	1.997	8.40E+10	1.35E+11
7	12:50-13:00	0.36	1985	5515	2.65E+08	7.35E+08	2.900	2.300	8.43E+10	1.43E+11
8	13:03-13:10	0.40	2143	5357	2.86E+08	7.14E+08	3.091	2.487	8.10E+10	1.42E+11
9	13:11-13:20	0.44	2292	5208	3.06E+08	6.94E+08	3.338	2.774	8.16E+10	1.48E+11
10	13:20-13:30	0.48	2432	5068	3.24E+08	6.76E+08	3.331	2.792	7.50E+10	1.39E+11

N2O bottle background:

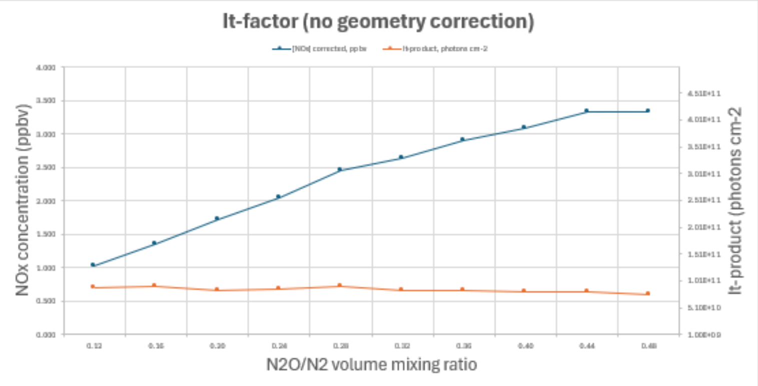
step	time (UTC+2)	$[NO_x]$ (ppbv)
1	08:05-08:16	0.113
2	08:17-08:25	0.131
3	08:25-08:37	0.250
4	08:38-08:45	0.249
5	08:45-08:55	0.259
6	08:55-09:05	0.264
7	09:05-09:15	0.221
8	09:15-09:25	0.225
9	09:25-09:35	0.185
10	09:35-09:45	0.160

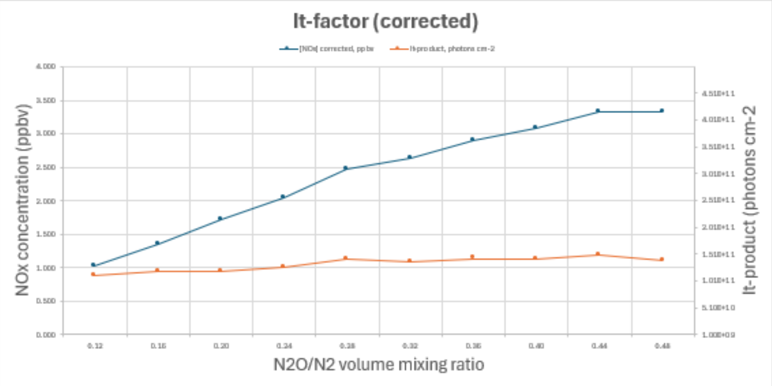
monitor background:

step	time (UTC+2)	$[NO_x]$ (ppbv)
1	07:55-08:05	0.379

UV-light attenuation correction:

step	volume mixing ratio N2O/N2	geometry correction factor K
1	0.12	0.80
2	0.16	0.75
3	0.20	0.71
4	0.24	0.67
5	0.28	0.64
6	0.32	0.62
7	0.36	0.59
8	0.40	0.57
9	0.44	0.55
10	0.48	0.54





RESULTS (@7.5 lpm):

It_mean	8.43E+10
It_median	8.43E+10
It_corrected_mean	1.32E+11
It_corrected_median	1.37E+11

RESULTS (@10 lpm):

It_mean	6.33E+10
It_median	6.33E+10
It_corrected_mean	9.93E+10
It_corrected_median	1.03E+11

Calculate the It-product

Option A - Using example_sheet.xlsx and It_factor_correction.ipynb

desired N_2O/N_2 mixing ratios for the actinometry (and background) experiment

start and end time of each step of the experiment

Info:

Date	dd/mm/yyyy
Calibrator	n
Operator	Name Surname
Email address	name.surname@helsinki.fi

It-product calculation:

parameter	value or expression
R1, σ_{N2O} (cm^2)	1.43E-19
R1, Φ_{N2O} (cm^2)	1.00E+00
R2, k_2 (cm^3 mol^{-1} s^{-1})	3.09E-11
R3, k_3 (cm^3 mol^{-1} s^{-1})	7.60E-11
R4, k_4 (cm^3 mol^{-1} s^{-1})	4.30E-11
R5, k_5 (cm^3 mol^{-1} s^{-1})	6.00E-12

T (K) 298.15

Actinometry experiment:

step	time (UTC+2)	volume mixing ratio N_2O/N_2	standard N_2O flow (sccm)	standard N_2 flow (sccm)	$[N_2O]$ (ppbv)	$[N_2]$ (ppbv)	$[NO_x]$ (ppbv)	$[NO_x]$ corrected (ppbv)	It-product (photons cm^{-2})	corrected It-product (photons cm^{-2})
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UV-light attenuation correction:

step	volume mixing ratio N2O/N2	geometry correction factor K
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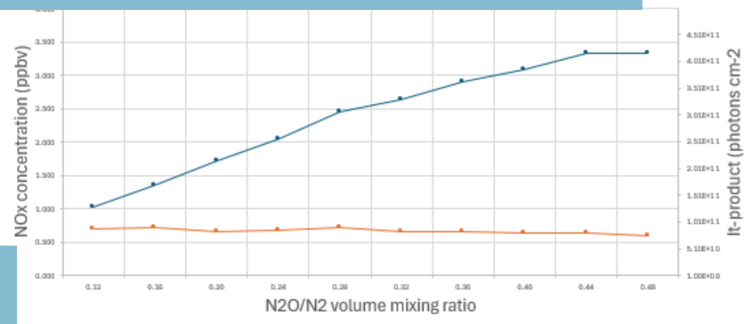
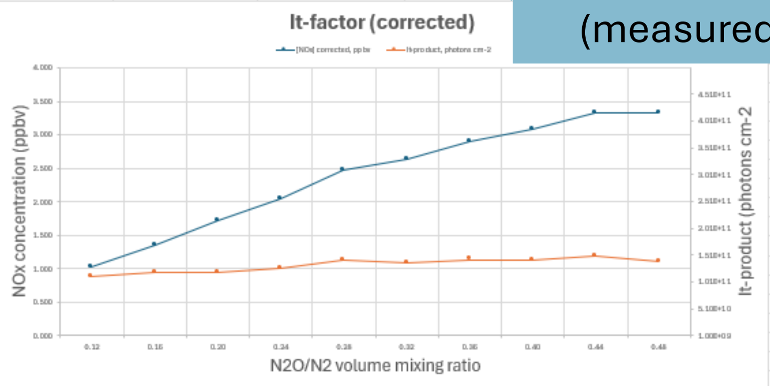
6	0.32	0.62
7	0.36	0.59
8	0.40	0.57
9	0.44	0.55
10	0.48	0.54

average temperature of N_2O and N_2 flows (displayed on the MFCs)

NO_x concentrations (measured by the monitor)





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It_corrected_median	1.03E+11

correction factor K for each N_2O/N_2 mixing ratio

Calculate the It-product

Option B - Using It_factor.m or It_factor.ipynb

❖ Adjust the following parameters in the user interface section (see Slide 18):

- *N_2O_flow*: N_2O flow rate (sccm)
- *N_2_flow*: N_2 flow rate (sccm)
- *NOx_ppb*: NO_x concentration measured by the NO_x monitor (ppb)
- *T*: Average temperature of the N_2O and N_2 flows (°C)
- *QtotNOx*: Total inlet flow rate used during the actinometry experiment (lpm)
- *Qinstrument*: Total inlet flow rate used during the sulfuric acid calibration experiment (lpm)
- *R*: Inner radius of the quartz tube (cm)
- *monitor_bg*: NO_x concentration (ppb) measured by the NO_x monitor when the N_2O flow is zero and the Hg lamp is on
- *bottle_bg*: NO_x concentration (ppb) measured by the NO_x monitor during the background experiment (Hg lamp off)

❖ The script outputs the mean and median values of both the uncorrected and geometry-corrected It-product for the total flow rate used during the sulfuric acid calibration experiment (the value specified for *Qinstrument*).

It_factor.m

It_factor.ipynb

Calculate the It-product

Option B - Using It_factor.m or It_factor.ipynb

It_factor.m

```matlab
%% N2O actinometry experiment | It-factor computation
% created by Lisa Beck
% modified by Cecilia Righi - latest update: July 2026

% --N2O_flow: set N2O flow rate (sccm)
% --N2_flow: set N2 flow rate (sccm)
% --NOx_ppb: NOₓ concentration detected by the NOₓ monitor (ppb)
% --T: average of N2O flow and N2 flow temperatures (°C)
% --QtotNOx: inlet flow rate used in the actinometry experiment (lpm)
% --Qinstrument: inlet flow rate used in the sulfuric acid calibration experiment (lpm)
% --R: inner radius of quartz tube (cm)
% --monitor_bg: NOₓ concentration (ppb) detected by the NOₓ monitor when
% N2O flow = 0 and Hg-lamp is on
% --bottle_bg: NOₓ concentration (ppb) detected by the NOₓ monitor during the background experiment (Hg-lamp off)


%% user interface
N2O_flow=[804 1034 1250 1452 1641 1818 1985 2143 2292 2432]; % sccm, ADJUST
N2_flow=[6696 6466 6250 6048 5859 5682 5515 5357 5208 5068]; % sccm, ADJUST
NOx_ppb=[4.325 5.650 6.749 7.642 8.609 9.410 10.060 10.657 11.241 11.581] ; % ppb, ADJUST

T=25; % °C, ADJUST

QtotNOx=7.5; % lpm, ADJUST
Qinstrument=7.5; % lpm, ADJUST

R=0.78; % cm, ADJUST

monitor_bg=0.8; % ppb, ADJUST
bottle_bg=[0.4 0.2 0.2 0.1 0.08 0.06 0.06 0.04 0.04 0.02];  % ppb, ADJUST
```

N2O actinometry experiment | It-factor computation

Last update: July 2026

```python
import pandas as pd
import datetime as dt
import numpy as np
import matplotlib.pyplot as plt
from datetime import timedelta
from scipy import integrate

##
# --N2O_flow: set N2O flow rate (sccm)
# --N2_flow: set N2 flow rate (sccm)
# --NOx_ppb: NOₓ concentration detected by the NOₓ monitor (ppb)
# --T: average of N2O flow and N2 flow temperatures (°C)
# --QtotNOx: inlet flow rate used in the actinometry experiment (lpm)
# --Qinstrument: inlet flow rate used in the sulfuric acid calibration experiment (lpm)
# --R: inner radius of quartz tube (cm)
# --monitor_bg: NOₓ concentration (ppb) detected by the NOₓ monitor when N2O flow = 0 and Hg-lamp is on
# --bottle_bg: NOₓ concentration (ppb) detected by the NOₓ monitor during the background experiment (Hg-lamp off)

##############################################################
# user interface
##############################################################

N2O_flow = np.array([804,1034,1250,1452,1641,1818,1985,2143,2292,2432]) # sccm, ADJUST
N2_flow = np.array([6696,6466,6250,6048,5859,5682,5515,5357,5208,5068]) # sccm, ADJUST
NOx_ppb = np.array([4.325,5.650,6.749,7.642,8.609,9.410,10.060,10.657,11.241,11.581]) # ppb, ADJUST

T = 25 # °C, ADJUST

QtotNOx=7.5 # lpm, ADJUST
Qinstrument = 7.5  # lpm, ADJUST

R = 0.78  # 0.78 cm, ADJUST

monitor_bg=0.8; # ppb, ADJUST
bottle_bg=[0.4,0.2,0.2,0.1,0.08,0.06,0.06,0.04,0.04,0.02];  # ppb, ADJUST
```

Calculate the It-product

Notes

- The values of K computed using any of the scripts are generally equivalent, with only negligible differences. These minor discrepancies arise from the use of different coordinate systems. *It_factor_correction.ipynb* and *It_factor.m* use cylindrical (polar) coordinates (r and φ), in which the optical path length, $L(r,\varphi)$, is expressed as:

$$L(r,\varphi) = \sqrt{R^2 - (r \cdot \sin\varphi)^2} + r \cdot \cos\varphi \qquad \text{(eq. 7)}$$

Hence, the integral used to calculate K follows the same formulation as Equation 4.

- Instead, *It_factor.ipynb* uses Cartesian coordinates (x and y), and the optical path length, $L(x,y)$, is written as:

$$L(x,y) = \sqrt{R^2 - y^2} + x \qquad \text{(eq. 8)}$$

This leads to an alternative formulation of the integral for K:

$$K = \frac{2\int_{-R}^{R}\int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} \exp\left(-\sigma_{N_2O}\cdot[N_2O]\cdot(\sqrt{R^2-y^2}+x)\right)\cdot dy\cdot dx}{\pi\cdot R^2} \qquad \text{(eq. 9)}$$

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